

2024



Modernizing Speech Data Collection Using AWS with Cerebral Palsy Alliance

Learn how Cerebral Palsy Alliance in healthcare built an innovative, gamified application for dysarthria research using AWS.

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Creates

a robust database of speech data

Connects

users and their families with researche



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[Cerebral Palsy Alliance](#) was looking to create a speech and facial-movement database, My Voice Library, of dysarthric voices for researchers and engineers to ultimately open opportunities for people living with cerebral palsy and other disabilities and their families. So, it turned to its partner [Kablamo](#) to build a gamified application using Amazon Web Services (AWS). The application would capture speech data through a series of interactive verbal exercises. Through its ongoing pilot project of My Voice Library, Cerebral Palsy Alliance is creating a robust database of speech data, with plans to provide children with dysarthria improved access to accurate treatment, to connect children with cerebral palsy and their families with researchers, and to open opportunities for future speech technology innovation.



Opportunity | Using AWS to Create a Voice Database for Cerebral Palsy Alliance

Cerebral Palsy Alliance is a groundbreaking global center of expertise for cerebral palsy research, advocacy, intervention, and assistive technology innovation. Its mission is to deliver world-class services, community advocacy and policy influence, training, global research, and advances in technology for people with cerebral palsy and similar disabilities. In 2021, researchers at Cerebral Palsy Alliance wanted to provide a platform to support the development of groundbreaking communication solutions for people with cerebral palsy and dysarthria, an endeavor that is currently challenged by the lack of large public human-voice datasets.

“We had this idea of creating a library of voices to provide a resource for developing near-real-time communication solutions,” says Dr. Petra Karlsson, technology program lead at Cerebral Palsy Alliance. “These solutions break down the barriers that people with cerebral palsy and dysarthria currently face when using assistive technology, such as taking 15–20 times longer to communicate than regular speech.”



To meet this need, Cerebral Palsy Alliance wanted to create an interactive, gamified experience where participants could contribute their voices to a database. So, Cerebral Palsy Alliance reached out to its partner Kablamo, an [AWS Partner](#) that focuses on building bespoke software solutions. Dr. Karlsson brought her concept for a gamified application to Kablamo, and the partner worked with her to create the game within budget constraints. Thus, My Voice Library was born. It is a user-friendly application for collecting speech data from children who have cerebral palsy and dysarthria. “

This project is a great example of the value that AWS brings to its customers and partners,” says Clare Burrows, partnerships and business development lead at Kablamo. “Being an AWS Partner gave us the opportunity to work with Cerebral Palsy Alliance to find the right solution.”

Solution | Building a Scalable, Cost-Effective Solution for Collecting Speech Data Using AWS

The initial discovery phase for My Voice Library kicked off in March 2021 and lasted 6 weeks. During that phase, the project team focused on the target audience of the application and decided how the application would be used. Overall, the approach was highly iterative, with Cerebral Palsy Alliance working closely with Kablamo to refine the final solution for good user experience. Then, the solution itself was built in 10 weeks and consisted of a simple yet effective backbone of AWS solutions.

For reliability and scalability, the team chose [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. To store the voice data, the team uses [Amazon Simple Storage Service](#) (Amazon S3), an object storage service offering cutting-edge scalability, data availability, security, and performance. For database and analytics needs, the team selected [Amazon Elastic Block Store](#) (Amazon EBS), an easy-to-use, scalable, high-performance block-storage service designed for Amazon EC2. “Scalability, accessibility, and cost optimization were the real drivers of how our solution was architected,” says Burrows. “Having a global, cloud-based solution on AWS fulfilled our needs for this project.”

My Voice Library provides an engaging and enjoyable way for children with dysarthria to share speech data. Players repeat sentences on camera, which then collects voice and facial-movement data. With various levels, a point system, and custom characters, players have flexibility and options on how they want to do the game. Each level is themed with fun facts and interesting trivia across a variety of topics. The application is accessible and reaches diverse players with its age- and gender-neutral content.



My Voice Library has begun as a pilot project for data collection and will reach about 100 users by 2025. Cerebral Palsy Alliance sees vast potential benefits of the application. For example, the organization envisions that My Voice Library can be a channel for parents of children with cerebral palsy to connect with researchers directly to find individual speech solutions. “We want our data to empower a whole international community of researchers and engineers to help people across the full spectrum of speech impairments,” says Karlsson. “The bigger picture is to eventually create bespoke solutions to meet every person’s need.”

As the team looks forward to the application’s potential impact on the lives of people with cerebral palsy living with dysarthria, it is extremely proud of what it has accomplished using AWS solutions. “There wasn’t a dry eye in the house when we presented this project at Cerebral Palsy Alliance showcase, and that comes from the passion we have to enact change and impact lives through our technology,” says Rochelle Pillari, partnership manager at Kablamo. “This has been a great collaboration that we are really proud of.”

Outcome | Opening Up Opportunities for Future Innovation

Going forward, Cerebral Palsy Alliance hopes My Voice Library will drive many future technology innovations in cerebral palsy research. The team also plans to incorporate AWS machine learning solutions in future phases of this project. “It’s been an absolute delight to work alongside AWS and Kablamo as one team on a mission to help everyone find their voice,” says Karlsson.

About Cerebral Palsy Alliance

Cerebral Palsy Alliance is a groundbreaking global center of expertise for cerebral palsy research, advocacy, intervention, and assistive technology innovation.

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Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

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Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry scalability, data availability, security, and performance.



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Amazon EBS

Amazon Elastic Block Store (Amazon EBS) is an easy-to-use, scalable, high-performance block-storage service designed for Amazon Elastic Compute Cloud (Amazon EC2).

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